

Round four, Rural R&D for Profit summary of approved projects

On 21 March 2019, the then Minister for Agriculture, Water and the Environment, approved funding of \$39.65 million from 2018-19 to 2021-22 for ten projects under round four of the Rural R&D for Profit program. This funding is matched by more than \$63 million in cash and in-kind contributions from research and development corporations and their project partners.

The successful projects will contribute to the program objective – to realise productivity and profitability improvements for primary producers.

The application period for the fourth round of funding opened on 10 August 2018 and closed 5 November 2018. There will be no further funding rounds.

Project title	Applicant	Grant funding awarded (GST excl)	Applicant/ partner contributions (GST excl.)	Contributors	Summary
Closing the Loop, Black Soldier Fly technology to convert agricultural waste into high quality fertiliser and soil improvers	Australian Pork Limited	\$1,712,000	\$856,000 cash \$1,599,278 in-kind	University of Western Australia, Dairy Australia, Fisheries Research & Development Corporation, Australian Eggs Limited, Future Green Solutions, Department of Agriculture and Fisheries Queensland, AgriFutures Australia, Australian Meat Processor Corporation	This project will investigate the development of Black Soldier Fly Farming (BSF) castings and larvae into high quality, low-cost, slow-release, granulated fertilizer products, that are safe to handle, transport and apply. The project will assist with overcoming existing adoption barriers by involving policy makers and farmers during field trials and assisting early adopters through extension activities. Adoption

Novel technologies and practises for the optimisation of pollination within protected cropping environments	Horticulture Innovation Australia Limited	\$4,318,514	\$2,175,757 cash \$2,385,723 in-kind	University of Adelaide, University of New England, University of Tasmania, Plant & Food Research Australia Pty Limited, NSW Department of Primary Industries	of BSF technology and its products has the potential to increase productivity and profitability via reduced input costs and generation of alternative revenue streams to a wide range of agricultural enterprises. This project will focus on ensuring the adoption of R&D outcomes for horticultural growers and apiarists through developing cross-commodity best practice resources that optimise the efficiency of the commercial hives and pollinators used for pollination services and improve the protected cropping systems. This will allow beekeepers to better prepare hives, companies to improve their technology, and growers to improve pollination under protected cropping environments to innovate the pollination space and further mature
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					effective mechanical pollination technology that builds a framework for integration into existing cropping systems.
					Smarter Irrigation for Profit Phase 2 project will address the challenge of reduced water availability by improving the water productivity of Australian cropping and pasture irrigators. It builds off the findings from the Smarter Irrigation for Profit Phase 1 project which found participating Australian irrigators could achieve a 10-30 percent improvement in water productivity by adopting precision irrigation technologies. This Phase 2 project incorporates three key areas of work:
Smarter Irrigation for Profit Phase 2	Cotton Research and Development Corporation	\$7,131,037	\$5,581,527 cash \$7,799,757 in-kind	Dairy Australia, Sugar Research Australia, AgriFutures Australia, Grains Research and Development Corporation, Tasmanian Institute of Agriculture, Deakin University, University of Melbourne, University of Southern Queensland, Agriculture Victoria, Gwydir Valley Irrigators Association, NSW Department of Primary Industries, CSIRO, University of Sydney	1. Developing new and innovative irrigation technologies, including the application of new sensors, advanced

analytics to improve irrigation scheduling and strategies to reduce water storage evaporation.

2. Optimising the benefits of automated irrigation for cotton, rice, sugar and dairy irrigation farming systems. Research will be conducted on commercial farms to enable farmer input and make sure outcomes are cost effective and practical.

3. Closing the irrigation productivity yield gap for cotton, rice, dairy, sugar and grains irrigators through best practice extension.

Honey bee genetic improvement program	Rural Industries Research and Development Corporation (AgriFutures Australia)	\$1,646,838	\$1,372,500 cash \$721,223 in-kind	AgriFutures Australia, NSW Department of Primary Industries, University of Sydney, Better Bees WA, When Bee Foundation, Commercial Beekeepers, Olam, Monson's Honey and Pollination, Costa Group,	The project will develop a national genetic improvement program to transform the performance of honey bees in Australia using innovative breeding technologies. The project will focus on selecting traits of importance to

						University of New England, Warren Taylor (Australian Queen Bee Exporters)	beekeeping, horticulture and broadacre industries dependent on honey bee pollination. A national database will be built to hold the honey bee genetic trait data that is collected. The project will also undertake an economic evaluation of breeding programs and develop and implement standardised selection methods that beekeepers can use to assess honey production, pollination and health traits.
Developing and commercialising advanced measurement technologies and feedback systems into globally competitive Australian meat value chains	Meat & Livestock Australia Limited	\$5,420,000	\$5,420,000 cash	\$8,293,427 in-kind		Australian Meat Processor Corporation, Australian Pork Limited, University of Adelaide, JBS Australia, Teys Australia, Woolworths, Frontmatec Group, Stockyard, WAMMCO, Fletcher International Exports, Murdoch University, University of New England, Gundagai Meat Processors,	This project will transform the global competitiveness of Australia's meat value chains by developing and commercialising advanced measurement technologies and creating feedback and decision-making systems linked to accurate measurement. The project aims to commercialise successful technologies that

Area Wide Management for cropping systems weeds, investigating the weed management, social and economic opportunity	Grains Research and Development Corporation	\$1,867,219	\$933,609.42 cash	Primary Industries and Regions South Australia, PorkScan Pty Ltd, Department of Primary Industries NSW, Department of Economic Development, Jobs, Transport and Resources Victoria, Department of Primary Industries and Regional Development WA, University of Melbourne, Thomas Foods International, University of Technology Sydney, Angus Australia, Scott Automation and Robotics, Coles	predict lean meat yield and eating quality, commercialise genetic tools, feedback and decision support systems using data stemming from these technologies and support the adoption and creation of systems for industry-wide auditing and calibration of new technologies.
				Cotton Research and Development Corporation, AgriFutures Australia, CSIRO, University of New South Wales, University of Queensland, University of Adelaide, Millmerran Landcare Group, Irrigation Research & Extension Committee Inc,	This project will identify the benefits, key principles and practices of successful weed Area Wide Management (AWM) by developing an improved understanding of the bio-physical, geographic, economic and social drivers of AWM success through studying key weed
			\$1,143,111.46 in-kind		

Underpinning agricultural productivity and biosecurity by weed biological control	Rural Industries Research and Development Corporation (AgriFutures Australia)	\$7,492,034	\$3,752,928 cash \$4,399,617 in-kind	Toowoomba Regional Council	species across diverse landscapes. The project will also characterise and identify the social and economic costs and benefits of weed management across a range of land uses.
				Grains Research and Development Corporation, Meat & Livestock Australia, Cotton Research and Development Corporation, CSIRO, NSW Department of Primary Industries, Queensland Department of Agriculture and Fisheries, Victoria Department of Economic Development, Jobs, Transport & Resources, AgForce, Bundaberg Regional Council, Castlereagh Macquarie County Council, Cramond Clan Environmental Consultants, Far North Queensland Regional Organisation of Councils, Gympie	This project will undertake foundational and applied research to contribute to integrated management of important weeds that affect Australian agriculture and water resources. The project will focus on 11 major weeds and complete risk assessments of promising biocontrol agents identified, undertake large-scale releases of approved agents and understand interactions between control methods, as well as progressing biocontrol research into new weed targets towards delivery. Field surveys will be used to validate a quantitative risk analysis approach to

				Regional Council, Hancock Queensland Plantations Pty Ltd, Hudson Pear Taskforce, Landcare Research, Malanda Beef Plan Group Inc, Mareeba Shire Council, NSW Environmental Trust, NSW Weed Biocontrol Taskforce, South East Queensland Water, South East Local Land Services NSW, Tablelands Regional Council, TasWeed Biocontrol, University of Queensland, University of Southern Queensland, US Department of Agriculture	determining candidate agents' suitability for release, providing more certainty around risk assessment data. The project will extend beyond the current project on weed biocontrol funded by the Rural R&D for Profit program to increase speed to impact.
Biorefineries for profit -Phase 2	Sugar Research Australia Limited	\$800,000	\$400,000 cash \$445,595 in- kind	Queensland University of Technology, Cotton Research and Development Corporation, Australian Pork Limited, Rivalea Australia, Mercurius Australia Pty Ltd, Bioproton Pty Ltd	The Biorefineries for Profitproject builds off the breakthroughs made in Phase 1 of the project funded under round one the Rural R&D for Profit program, which developed and demonstrated a range of innovative biorefinery

technologies that convert Australian agricultural and forestry feedstocks into new value-added animal feeds, chemicals, and advanced fuels. This Phase 2 project will progress the commercialisation of the biorefinery technologies. It will demonstrate the production of (a) highly digestible animal feed from bagasse at pilot-scale and undertake feeding trials, (b) probiotic feed supplements and demonstrate efficacy in feeding trials, and (c) a chemical (5-chloromethyl furfural, CMF) from cotton gin trash at pilot-scale.

Boosting diagnostic capacity for plant production industries

Grains Research and Development Corporation

\$4,600,000

\$3,083,805 cash

\$7,142,329 in-kind

Horticulture Innovation Australia, Sugar Research Australia, Cotton Research and Development Corporation, Wine Australia, Forests and Wood Products Australia, Agriculture Victoria, South Australia Research and Development

This project will provide a platform to develop and implement improved national diagnostics for key biosecurity threats by developing diagnostics that facilitate early and accurate diagnosis and rapid response to threats that impact on

				Institute, Queensland Department of Agriculture and Fisheries, AgriFutures Australia, Plant Health Australia, NSW Department of Primary Industries, NT Department of Primary Industry and Resources, WA Department of Primary Industries and Regional Development, CSIRO, Tasmania Department of Primary Industries, Parks, Water and Environment, Plant and Food Research New Zealand, Cesar Pty Ltd	production and access to domestic and international markets. This will be achieved through capacity building for industry for early detection of pests and improved sensitivity of diagnostics. Industry and government responses to key threats will be tested with an aim to create maximum value to industry through better partnerships in diagnostic decision- making for biosecurity response efforts.
Multi-scale Monitoring Tools for Managing Australian Tree Crops - Phase 2	Horticulture Innovation Australia Limited	\$4,660,821	\$3,067,045 cash \$2,745,023 in- kind	Ceres Imaging, Tie Up Farming, NSW Department of Primary Industries, University of New England, NT Department of Primary Industry and Resources, Australian Mango Industry Association, Queensland Department of	This project is a continuation from the Phase 1 project that identified a range of emerging technologies that supports more accurate yield and fruit quality forecasting and mapping, improved monitoring of stresses at the individual tree and orchard level and

		provides the industry with an accurate understanding of the extent and distribution of production as well as supporting improved biosecurity response and post disaster response. Phase 2 of project builds on the foundations established throughout the first stage by expanding and encompassing additional industries, new partners and technologies with a focus on extension and adoption. The project directly responds to the specific ‘needs’ of the participating industries including more accurate pre-harvest yield forecasting and yield mapping; improved monitoring of the fruit quality and tree health, including biosecurity threats; and a national map of commercial orchards.
	Agriculture and Fisheries,Central Queensland University	
		\$26,643,171
Totals		\$39,648,463 cash

\$36,675,083
in-kind

\$63,318,254
total

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/innovation/rural-research-development-for-profit/approved-projects-round4>